

Oliver Ye

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EDUCATION

Yale University | BS in Computer Science, GPA: 3.8

New Haven, CT

Class of 2024

- **Relevant Coursework:** Robotics Modeling & Control, Applied Planning and Optimization, Mobile and Embedded Systems, Networks, Computer Architecture, Parallel Programming, Operating Systems, Algorithms

EXPERIENCE

Applied Intuition | Embedded ML & Firmware Engineer

August 2024 – Present

Mountain View, California

- Optimized deformable-agg CUDA kernel via shared memory and ILP, cutting latency 28% (44% throughput gain)
- Eliminated GPU contention via hardware accelerator offloads, reducing system latency an additional 24%
- Built model optimization pipeline (quantization, FX graph surgery) enabling compilation of custom models across 3 product verticals and 6 HW platforms with varying GPU architectures
- Enabled strongly-typed Dynamo tracing for model export, auditing and resolving data-dependent control flow to cut intermediate-tensor error from 8% to 0.01% on FP32 and stabilize a 60%-flaky export process
- Implemented network stack from Ethernet drivers to protocol (gPTP, AVTP, TCP/IP) spanning 10+ HW platforms for RTOS-based zonal computes, debugged DMA race conditions, raising car uptime/availability from 1 to 62 hrs
- Maintained HIL infrastructure and contributed to 15+ custom benches enabling remote development
- Contributed to NVIDIA DriveOS bring-up, running QNX as a hypervisor guest OS on custom board

Amazon Robotics | Software Development Intern

June 2023 – August 2023

Westborough, Massachusetts

- Implemented optimized spline-based trajectory generation on warehouse robot
- Profiled trajectory generation execution time scaling as a function of waypoint counts

Zipline | Embedded Systems Intern

May 2022 – August 2022

San Francisco, California

- Prototyped over-the-air updating system for Linux distribution on Jetson Xavier NX SOM
- Implemented protection from power cycle, repartitioning of disk, and boot-chain modifications on Yocto Linux image

Yale Bulldogs Racing | President & Electrical Systems Lead

October 2020 – August 2024

New Haven, Connecticut

- Implemented low voltage communication systems and designed new high voltage battery pack
- Mentored and organized sub-teams of new members to develop problem solving and engineering skills

Yale Bhattacharjee Labs | Undergraduate Researcher

May 2021 – December 2021

New Haven, Connecticut

- Utilized High Level Synthesis tools for power-optimized system architecture of Brain-Computer Interfaces

SIDE PROJECTS

Pocket Player — TinyML Audio Device

2026 – Present

- Built ESP32 music player with BT/wired audio; planned custom PCB and on-device ML for gesture control

Out-of-Order Processor in Verilog

2023

- Implemented out-of-order, superscalar processor in Verilog for toy ISA

Autonomous Underwater Vehicle (AUV)

2023

- Prototyped and designed basic AUV from scratch with 3D printed parts

SKILLS AND INTERESTS

Programming Languages: C, C++, ARM Assembly, Python, CUDA, Rust, Verilog

Systems: Bare Metal, RTOS, Embedded Linux

Hardware: ARM Cortex M/R cores, NXP SoC, NVIDIA Orin, STM32

Interests: Model compression (sparsity, INT4/INT8/FP8 quantization, distillation, kernel fusion); computer architecture (cache hierarchies, memory bandwidth, SIMD/vectorization, RISC-V, GPU, NPU/accelerators); HW/ISA for ML (ARM Ethos, SME/SVE), MLIR